

Haoran Liang (梁浩然)

Email: haoran.1.liang@kcl.ac.uk

Website: hrliangmath.github.io

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Education

2024.06 –	Mathematics PhD , King's College London (aligned with LSGNT)	London, UK
	advisors: Prof. Payman Kassaei and Prof. Fred Diamond, potential thesis topics: differential operators on p -adic automorphic forms (group) mini project: The eigencurve at classical weight 1 points (with Lucie Gatzmaga), supervised by Dr. George Boxer	
2023.10 – 2023.12	Visiting student, Morningside Center of Mathematics	Beijing, China
2021.09 – 2023.10	Mathematics MSc. (Dist) , ETH Zürich	Zürich, Switzerland
	advisor: Prof. Emmanuel Kowalski, thesis: automorphic forms on adèle groups	
2020.01 – 2020.05	Visiting student, UC Berkeley	California, USA
2018.07 – 2018.08	Visiting student, University of Alberta	Alberta, Canada
	Excellence (group mini project on discrete dynamics)	
2017.09 – 2021.06	Mathematics BSc. (Hons) , Xi'an Jiaotong University	Xi'an, China
	thesis: (Algebraic) number fields and special cases of Fermat's last theorem, A+ supervised by: Prof. Ping Xi	

Academic Experiences

• Invited Talks

2026.10	What is... the dual BGG complex? Zürich Graduate Colloquium
2026.09	p -Adic families of automorphic forms, Y-RANT VIII
2026.05	Theta operators on (p -adic) automorphic forms and applications, Oxford Junior Number Theory Seminar
2026.03	Automorphic forms as bridge and gap: a tale of two problems, London Junior Number Theory Seminar
2025.09	Local geometry of the eigencurve, Y-RANT VII
2025.07	Local geometry of the eigencurve, XJTU number theory seminar
2025.01	O-minimality and Diophantine applications, London Junior Number Theory Seminar
2023.09	Automorphic forms on adèle groups, XJTU basic notions seminar
2023.09	O-minimality and Diophantine applications, XJTU basic notions seminar

• Teaching assistance

2025.09 – 2025.12	Groups and Symmetries WS24/25, KCL (Dr. Paul Cook)
2024.09 – 2024.12	Advanced Algebra WS24/25, KCL (Dr. Alex Torzewski)
2024.09 – 2024.12	Basic Algebra WS24/25, KCL (Prof. Beth Romano)
2022.09 – 2022.12	Complex Analysis WS22/23, ETH Zürich (Prof. Emmanuel Kowalski)
2020.09 – 2021.01	Modern Algebra WS20/21, XJTU (Prof. Ping Xi)

- **Semester / course projects**

- 2022.10 – 2023.01 Modular curves and dimension formulae for certain spaces of modular forms
supervised by: Prof. Sarah Zerbes
- 2022.03 – 2022.06 The class number formula and beyond
supervised by: Prof. Claire Burrin
- 2022.07 Schoof’s algorithm (for counting points on elliptic curves over finite fields)
course project for *Computations in Algebra & Arithmetic* by Prof. David Loeffler
- 2020.12 – 2021.02 A glimpse at classical algebraic geometry
supervised by Prof. Ping Xi
- 2020.01 – 2020.05 A Journey through combinatorics
(*directed reading program* at UC Berkeley) directed by: Yulia Alexandr
- 2019.06 – 2019.09 A unified approach for constructing cospectral (signed) graphs
supervised by: Prof. Wei Wang

- **Conferences, summer schools, and workshops attended**

- 2026.09 [Mod p Hecke Algebras and Galois Moduli](#), Bergische Universität Wuppertal
- 2026.09 [Young Researchers in Algebraic Number Theory VIII](#), University of Bristol
- 2026.07 [LMS-Sheffield School on the Langlands Program](#), University of Sheffield
- 2026.06 [Classical and p-Adic Aspects of the Kudla Program](#), CIRM
- 2025.10 [New Structures and Techniques in p-adic Geometry](#), IHES
- 2025.09 [Young Researchers in Algebraic Number Theory VII](#), University of Nottingham
- 2025.03 [Arizona Winter School 2025: Representation of p-adic Groups](#), University of Arizona, Tucson
- 2024.09 [School on Hodge Theory and Shimura Varieties](#), Universität Duisburg-Essen
- 2024.09 [Representations of p-adic Groups and the Langlands correspondence, in honor of Colin Bushnell](#), King’s College London
- 2024.08 [Young Researchers in Algebraic Number Theory VI](#), University of Oxford
- 2024.04 [2024 Jinling Number Theory and arithmetic Geometry Conference](#), Nanjing University
- 2023.11 [Langlands Correspondence, Automorphic Representations, and Related Topics](#), Wuhan University
- 2023.06 [Workshop on “Arithmetic Statistics in Automorphic Forms and Analytic Number Theory”](#), EPF Lausanne
- 2023.05 [Summer School on High-dimensional Expanders](#), Ghent University
- 2022.08 [Modular Forms in Number Theory and Beyond](#), Universität Bielefeld
- 2021.07 [Number Theory Summer School](#), Xi’an Jiaotong University
- 2020.08 [Summer School on Differential Geometry](#), Peking University

Extracurricular Activities

- 2020.09 – 2021.06 Student Association of Cultural & Historical Relic Fans, XJTU
- 2017.09 – 2020.06 Fenghuang Student Reading Club, XJTU